

Full Markdown Functionality Reference

This reference covers all 20 feature categories:

- headings (with custom IDs),
- inline styling (bold/italic/strikethrough/highlight/sub/sup/underline/code),
- links and images,
- fenced code blocks,
- horizontal rules,
- lists (unordered/ordered/task),
- blockquotes,
- tables (with alignment and table foot),
- definition lists,
- footnotes,
- language markers,
- ruby annotations,
- emoji shortcodes,
- typography/legal marks,
- hard line breaks,
- HTML comments,
- YAML front matter,
- predefined CSS styles,
- backslash escaping,
- and paragraph handling.

Programmatic Usage

```
from markdown2html5_base import MarkdownToHTML
converter = MarkdownToHTML()
html = converter.convert("# Hello")
print(html)
```

1. Headings (H1-H6)

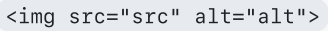
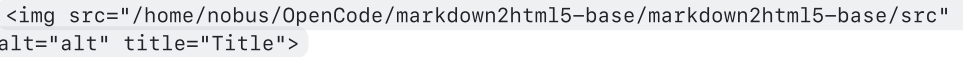
Markdown	Output HTML
# Heading 1	Heading 1
## Heading 2	Heading 2
### Heading 3	Heading 3
#### Heading 4	Heading 4
##### Heading 5	Heading 5
##### Heading 6	Heading 6

Custom ID: ## Section {#sec1} => <h2 id="sec1">Section</h2>

2. Inline Text Styling

Markdown	Output HTML
bold italic	bold italic
__bold italic__	bold italic
bold	bold
__bold__	bold
italic	italic
italic	italic
~~strikethrough~~	strikethrough
==highlight==	<mark>highlight</mark>
~subscript~	_{subscript}
^^underline^^	<u>underline</u>
^superscript^	^{superscript}
`code`	<code>code</code>

3. Links and Images

Markdown	Output HTML
[text](url)	text
![alt](src)	
![alt](src "Title")	

4. Fenced Code Blocks

A language tag after the opening fence (three backticks before and after the code) is rendered as a `<div class="code-lang">/python/</div>` label above the `<code>` element:

```
<div class="code-lang">&sol;python&sol;</div><pre><code>def hello():
    print("Hi")</code></pre>
```

Without a language tag, the code renders plainly as `<pre><code>` .

5. Horizontal Rules

`---` / `***` / `___` (3+ chars) => `<hr>`

6. Lists

1. **Unordered:** `* Item` or `- Item` => `<ul><li>Item</li></ul>`

2. **Ordered:** `1. Item` => `<ol><li>Item</li></ol>`

3. **Task:**

`* [ ] todo` => `<li><input type="checkbox" disabled> todo</li>`

`* [x] done` => `<li><input type="checkbox" checked disabled> done</li>`

7. Blockquotes

`> text` => `<blockquote><p>text</p></blockquote>` : blank lines within a blockquote split into separate `<p>` tags.

8. Tables

Supports `<thead>` , `<tbody>` , `<tfoot>` , and alignment:

Separator	Alignment
<code>:---</code>	left
<code>:---:</code>	center
<code>---:</code>	right

Footer: a row of `=` signs in the separator columns after body rows renders `<tfoot>` .

```
| Product | Qty | Price |
| :----- | :-: | ----: |
| Apples  | 2   | $3.00 |
| Bananas | 3   | $1.50 |
| Cherries| 1   | $4.00 |
|=====|=====|=====|
| Total   | 6   | $8.50 |
```

=>

```

<table>
  <thead>
    <tr>
      <th style="text-align:left;">Product</th>
      <th style="text-align:center;">Qty</th>
      <th style="text-align:right;">Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td style="text-align:left;">Apples</td>
      <td style="text-align:center;">2</td>
      <td style="text-align:right;">$3.00</td>
    </tr>
    <tr>
      <td style="text-align:left;">Bananas</td>
      <td style="text-align:center;">3</td>
      <td style="text-align:right;">$1.50</td>
    </tr>
    <tr>
      <td style="text-align:left;">Cherries</td>
      <td style="text-align:center;">1</td>
      <td style="text-align:right;">$4.00</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td style="text-align:left;">Total</td>
      <td style="text-align:center;">6</td>
      <td style="text-align:right;">$8.50</td>
    </tr>
  </tfoot>
</table>

```

9. Definition Lists

Term
 : Definition 1
 : Definition 2

=>

```

<dl>
  <dt>Term</dt>
  <dd>Definition 1</dd>
  <dd>Definition 2</dd>
</dl>

```

10. Footnotes

Reference: `[^1]` => `<sup id="fnref:1"><a href="#fn:1" class="footnote-ref">1</a></sup>`

Definition: `[^1]: Text` at bottom => rendered in `<div class="footnotes"><ol>...</ol></div>`

11. Language Markers

Annotate blocks or inline text with a language using a valid BCP 47 tag (e.g. `de`, `fr`, `zh-Hans`).

- **Block level:** Place `{:lang}` (with a space after it) at the very start of a line, before the Markdown tag. It renders as the global `lang` attribute on the resulting element and needs no closing marker:

```
{:de} # Überschrift
{:fr} Un paragraphe français.
{:ru} - Пункт списка
{:de} > Ein Zitat
{:de} | Kopf | Kopf |
      | ---- | ---- |
      | Zelle | Zelle |
```

=>

```
<h1 lang="de">Überschrift</h1>
<p lang="fr">Un paragraphe français.</p>
<ul lang="ru">
  <li>Пункт списка</li>
</ul>
<blockquote lang="de">...</blockquote>
<table lang="de">...</table>
```

- **Inline level:** Wrap text with `{:lang}...{:}` to render a `<span lang="...">`:

A French phrase `{:fr}"L'État c'est moi"{:}` is traditionally attributed to King Louis XIV of France.

=>

```
<p>A French phrase <span lang="fr">"L'État c'est moi"</span> is traditionally attributed to King Louis XIV of France</p>
```

12. Ruby Annotations

`{日本語|にほんご}` => `<ruby>日本語<rp>( </rp><rt>にほんご</rt><rp>)</rp></ruby>`

13. Emoji Shortcodes

- `:joy:` 😄
- `:smile:` 😊
- `:heart:` ❤️
- `:thumbsup:` 👍
- `:thumbsdown:` 👎
- `:wink:` 😉
- `:tada:` 🎉
- `:rocket:` 🚀
- `:fire:` 🔥
- `:star:` ★
- `:cry:` 😭
- `:thinking:` 🤔
- `:100:` 💯

- `:sparkles:` ✨
- `:eyes:` 👁️
- `:bulb:` 💡
- `:warning:` ⚠️
- `:ok:` 🍷
- `:check_mark:` ✓

14. Typography and Legal Marks

Name	Input	HTML Output
Copyright	<code>(c)</code>	<code>&copy;</code>
Trademark	<code>(tm)</code>	<code>&trade;</code>
Registered Trademark	<code>(r)</code>	<code>&reg;</code>
Plus-Minus	<code>+/-</code>	<code>&plusmn;</code>
Not Equal To	<code>!=</code>	<code>&ne;</code>
Logical Equivalence	<code><=></code>	<code>&hArr;</code>
Less-Than or Equal To	<code><=</code>	<code>&le;</code>
Greater-Than or Equal To	<code>>=</code>	<code>&ge;</code>
Left Arrow	<code>-></code>	<code>&rarr;</code>
Right Arrow	<code><-</code>	<code>&larr;</code>
Up Arrow	<code>:uparrow:</code>	<code>&uarr;</code>
Down Arrow	<code>:dnarrow:</code>	<code>&darr;</code>
Logical Implication	<code>=></code>	<code>&rArr;</code>
One-Half	<code>1/2</code>	<code>&frac12;</code>
One-Third	<code>1/3</code>	<code>&frac13;</code>
Two-Thirds	<code>2/3</code>	<code>&frac23;</code>
One-Quarter	<code>1/4</code>	<code>&frac14;</code>
Three-Quarters	<code>3/4</code>	<code>&frac34;</code>
Solidus (Slash)	<code>:slash:</code>	<code>&sol;</code>
Reverse Solidus (Backslash)	<code>:bslash:</code>	<code>&bsol;</code>
Left Angle Quote	<code><<</code>	<code>&laquo;</code>
Right Angle Quote	<code>>></code>	<code>&raquo;</code>
Smart Double Quotes	<code>"text"</code>	<code>&ldquo;text&rdquo;</code>
Smart Single Quotes	<code>'text'</code>	<code>&lsquo;text&rsquo;</code>
Straight Apostrophe	<code>'</code>	<code>&apos;</code>
Em Dash	<code>---</code>	<code>&mdash;</code>
En Dash	<code>--</code>	<code>&ndash;</code>
Ellipsis	<code>...</code>	<code>&hellip;</code>

15. Hard Line Breaks

End line with two spaces or backslash: `<br />`

16. HTML Comments

[comment]: # => `<!--comment-->`

17. YAML Front Matter

If the file begins with a YAML front matter block between `---` lines, the converter emits a complete HTML5 document with the metadata in `<head>` and the body content between `<body>` tags. Without front matter, the output stays a bare HTML fragment. The `--css` option (or `include_css=True`) embeds the default `<style>` block regardless of whether front matter is present.

```
---
lang: en
title: My Document
author: Jane Doe
description: A short description.
keywords: python, markdown, html5
published: 2026-08-09
---
```

- `lang` becomes the `<html lang="...">` attribute (a valid BCP 47 tag).
- `title` becomes the `<title>` element.
- `author`, `description`, `keywords`, and `published` become `<meta name="..." content="..." />` tags.
- A default `<style>` block with viewing-friendly CSS is embedded in `<head>`.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="author" content="Jane Doe" />
    <meta name="description" content="A short description." />
    <meta name="keywords" content="python, markdown, html5" />
    <title>My Document</title>
    <meta name="published" content="2026-08-09" />
    <style>
      /* default viewing-friendly CSS */
    </style>
  </head>
  <body>
    ...
  </body>
</html>
```

Any other keys are ignored, and if the file has no front matter at all, the output remains a bare fragment (unless `--css / include_css=True` is used, in which case it becomes a full document).

18. Backslash Escaping

Escape any special char: `\*`, `\#`, `\[`, etc.

Escapable: `\`*_{}[]()#+-.!|~^=:<>`

19. Paragraphs

Consecutive text lines merge into `<p>` . Blank lines separate paragraphs.

Empty line after list close => `<!-- -->` preserves whitespace.

20. CSS Styles

The converter (with `--css` option or `include_css=True`) embeds a default `<style>` block in `<head>` that provides viewing-friendly styling, regardless of YAML front matter. This predefined CSS rules includes:


```

body {
  padding: 20px;
  font-family:
    "Noto Serif",
    "Liberation Serif",
    "Times New Roman",
    Times,
    serif;
  font-size: 18px;
  line-height: 1.4;
  color: #000000;
}
h1, h2, h3, h4, h5, h6 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family:
    "Noto Sans",
    "Liberation Sans",
    Arial,
    sans-serif;
  font-weight: bold;
}
h1 { font-size: 32px; }
h2 { font-size: 28px; }
h3 { font-size: 24px; }
h4 { font-size: 20px; }
h5 { font-size: 18px; }
h6 {
  font-size: 18px;
  font-style: italic;
}
hr {
  height: 4px;
  margin: 20px 0;
  border: none;
  background-color: #000000;
}
li {
  position: relative;
  padding-left: 20px;
}
dt { font-weight: bold; }
dd {
  position: relative;
  margin-left: 0;
  padding-left: 20px;
  font-style: italic;
}
blockquote {
  margin-left: 0;
  padding-left: 20px;
  border-left: 8px solid #f5f5f5;
}
mark {
  padding: 0 2px;
  border-radius: 4px;
  background-color: #ffff00;
}
a:link { color: #0000cd; }
a:visited { color: #9400d3; }

```

```

a:hover, a:focus {
  outline: none;
  color: #000080;
}
a:active { color: #dc143c; }
code {
  padding: 2px 4px;
  border-radius: 4px;
  font-family:
    "Noto Sans Mono",
    "Liberation Mono",
    "Courier New",
    Courier,
    monospace;
  font-size: 0.9em;
  line-height: 1;
}
pre {
  max-width: 100%;
  margin: 0;
  padding: 20px;
  border: 1px solid #000000;
  background-color: #f5f5f5;
  overflow: auto;
  scrollbar-color: #000000 transparent;
}
pre > code {
  display: block;
  margin: 0;
  padding: 0;
  border: none;
  border-radius: 0;
  line-height: 1.2;
  background-color: transparent;
  overflow: visible;
}
div.code-lang {
  display: block;
  padding: 10px 20px;
  font-family:
    "Noto Sans Mono",
    "Liberation Mono",
    "Courier New",
    Courier,
    monospace;
  font-size: 0.9em;
  line-height: 1;
  background-color: #000000;
  color: #ffffff;
  font-weight: bold;
}
table {
  margin: 20px 0;
  border-collapse: collapse;
}
th, td {
  padding: 10px 12px;
  border: 1px solid #000000;
}
th { font-weight: bold; }
thead tr {

```

```
background-color: #000000;
color: #ffffff;
}
tfoot tr {
background-color: #f5f5f5;
font-style: italic;
}
ruby { ruby-position: over; }
rt {
letter-spacing: 0.05em;
font-size: 0.55em;
line-break: strict;
}
rp { display: none; }
span[lang="ja"] {
font-family:
"Noto Serif CJK JP",
"Source Han Serif JP",
"源ノ明朝",
"Source Han Serif",
"Hiragino Mincho ProN",
"Hiragino Mincho Pro",
"IPAexMincho",
"IPAMincho",
"MS PMincho",
"MS Mincho",
serif;
}
span[lang="zh-CN"], span[lang="zh-Hans"] {
font-family:
"Noto Serif CJK SC",
"Source Han Serif SC",
"思源宋体",
"Source Han Serif CN",
"Source Han Serif",
"Songti SC",
"FandolSong",
"WenQuanYi Bitmap Song",
"SimSun",
serif;
}
span[lang="zh-TW"], span[lang="zh-Hant"] {
font-family:
"Noto Serif CJK TC",
"Source Han Serif TC",
"思源宋體",
"Source Han Serif TW",
"Source Han Serif",
"Apple LiSung",
"LiSong Pro",
"HanaMinA",
"PMingLiU",
"MingLiU",
serif;
}
span[lang="zh-HK"] {
font-family:
"Noto Serif CJK HK",
"Source Han Serif HK",
"思源宋體 香港",
"思源宋體",
```

```
"Source Han Serif",  
"Apple LiSung",  
"LiSong Pro",  
"HanaMinA",  
"MingLiU_HKSCS",  
"PMingLiU",  
"MingLiU",  
serif;  
}  
span[lang="ko"] {  
  font-family:  
    "Noto Serif CJK KR",  
    "Source Han Serif KR",  
    "본명조",  
    "Source Han Serif",  
    "AppleMyungjo",  
    "UnBatang",  
    "은바탕",  
    "Batang",  
    serif;  
}
```